

OPERATION MANUAL

FOR

STI CB-15

CAPSULE BANDING MACHINE



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1. Specification

- 1.1 Name : STI Capsule Banding Machine
- 1.2 Model : CB-15
- 1.3 Scope of usage : The apparatus is used to band the joint portions of the cap and body of filled hard gelatin capsules.
- 1.4 Utilized capsule : Applicable to Elanco hard gelatin capsules size 000, 00, 0, 1, 2, 3, and 4
- 1.5 Electricity : Single phase 115 Volts 60 cycles
- 1.6 Capacity : Maximum 15,000 capsules/hour
- 1.7 Installation conditions: **Install in clean room with air control of 24 to 26°C and relative humidity below 55%. Room floor must be level.**
- 1.8 Required facility: Compressed air 5kg/cm 11 CFM
Vacuum 2000mmAq 1.7m/min
- 1.9 Main body equipment:
- | | |
|----------------------------|--------------------------------|
| Main motor | AC115V (Speed control motor) |
| Brush Roller motor | 24VoltDC (Speed control motor) |
| Banding roller motor | AC115V (Speed control motor) |
| Solution circulating motor | 24VoltDC |
| Banding pan heater | AC115V (Panel heater) |
| Banding tank heater | AC115V (Panel heater) |
| Drying fan | AC115V 7 total |
- 1.10 Dimensions:
- | | |
|-----------------|---|
| Banding machine | Width 2000 x Length 850 x Height 1550mm |
|-----------------|---|
- 1.11 Operational Vacuum Pump: Special voltage on request

SCHAEFER TECHNOLOGIES, INC.

STI CB-15

CAPSULE BANDING MACHINE

USER'S MANUAL

INTRODUCTION AND GENERAL RECOMMENDATIONS

INTRODUCTION

This User's Manual provides the user with information about how to use and maintain the machine correctly. It is compulsory to read this manual. It is our intention that the information contained in this manual will be a valid aid for your work. This information is based on data and on our best current knowledge. Carefully read this manual, including the recommendations and suggestions. No part of this manual may be copied or passed to third parties without the prior written consent of Schaefer Technologies, Inc.

We advise you to carefully read this manual, as it is essential to install, check and service the machine correctly in order to keep it in a perfectly efficient working order. The manual contains tables, designs and diagrams, which will help you to get to know the machine in detail. Please don't hesitate to contact us if you find that an explanation has been omitted or is unclear.

NAMEPLATE

For any servicing or overhaul operations not included in this manual or for any other technical problem which may arise, our Technical Personnel are available to provide you with information and to discuss any necessary action to take. When contacting us, please indicate the data on the machine's nameplate.

PRELIMINARY REMARKS

- The drawings and designs which show the system should be considered as reference and not necessarily faithful in every detail.
- The system dimensions and specifications contained in this manual are not binding and may be changed without warning.
- The designs and all of the other documents supplied as part of the system are the property of Schaefer Technologies, Inc.

The system is covered by a guarantee and a contract of purchase. During the period in which the machine is under guarantee, any repairs that have not been authorized by Schaefer Technologies, Inc. and are carried out on the same machine will automatically make the warranty void.

The user must take particular note of the safety warnings and information indicated by the following symbols and labels in this manual.



Mandatory



Danger



Lockout



Prohibited



Read manual



Wear eye protection



Wear safety gloves

115 VOLTS

Electrical Voltage Specification. 115 Volts



Electrical Current Specification. Single Phase



Caution – Pinch Point



Caution – Pinch point hazard. Keep hands clear.

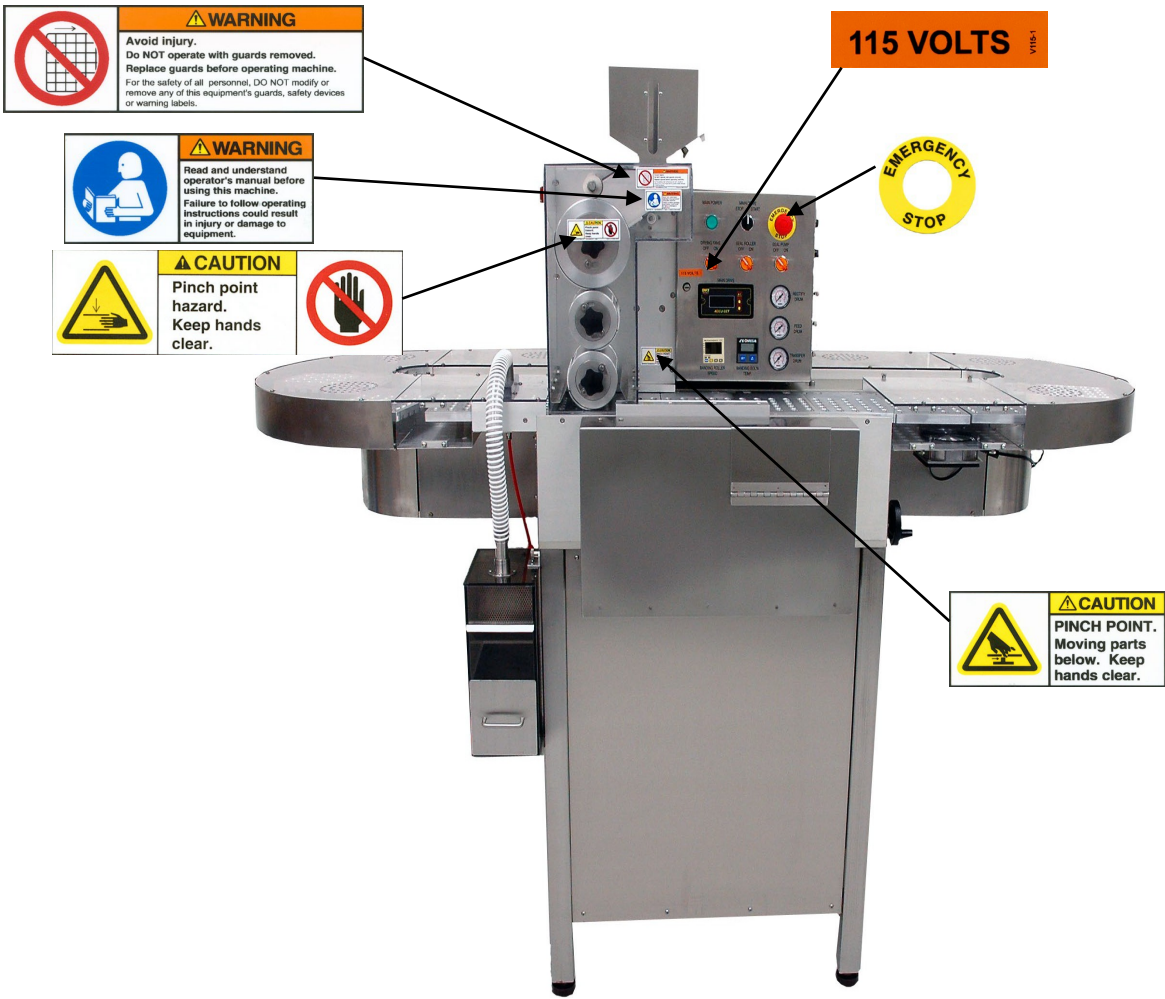


Warning – Avoid Burns. Do NOT touch hot surface.

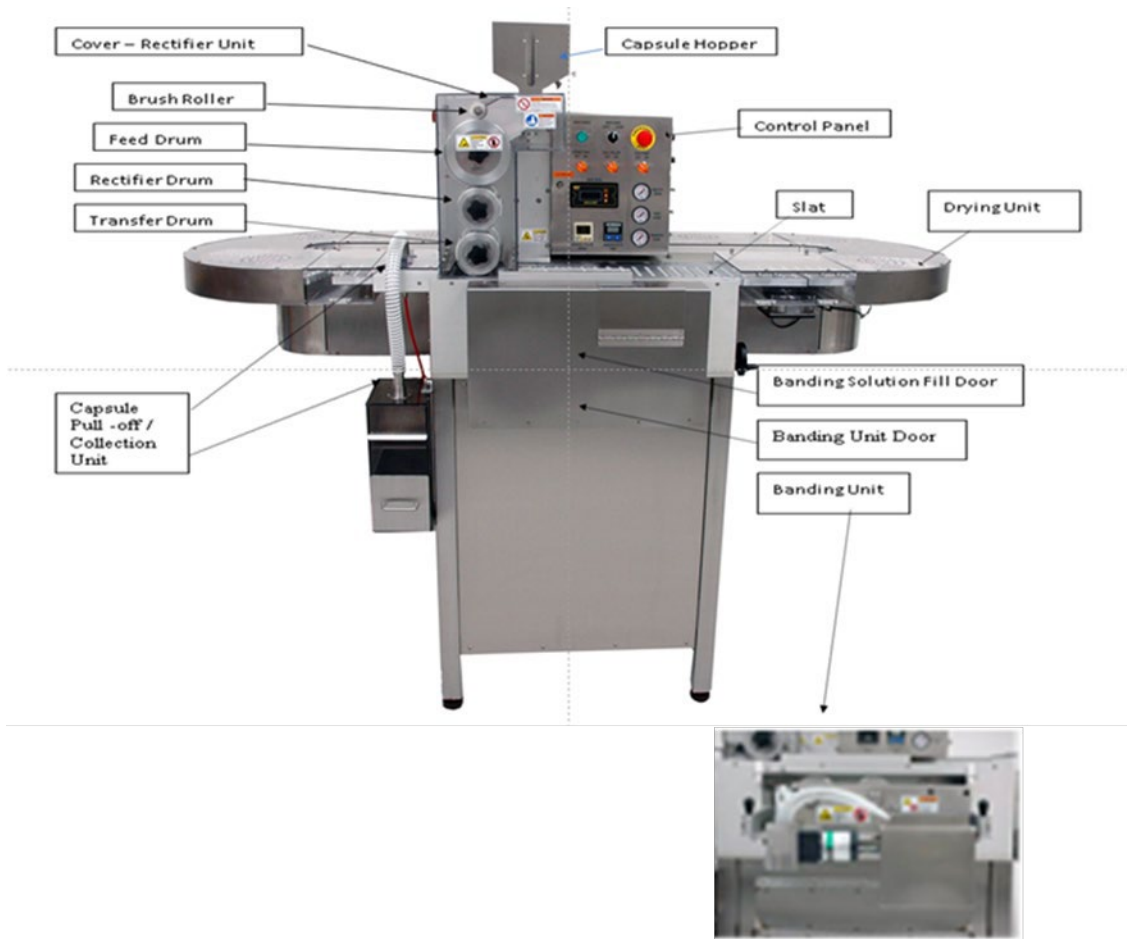


**Warning – Pinch Point. Keep hands clear of Roller/Belts.
Do not operate with open cover.**

WARNING
 Read and understand operation manual prior to operating or servicing this machine. Failure to follow operating instructions could result in personal



2. Name of each section



3. Banding Operation

3.1. Preparations

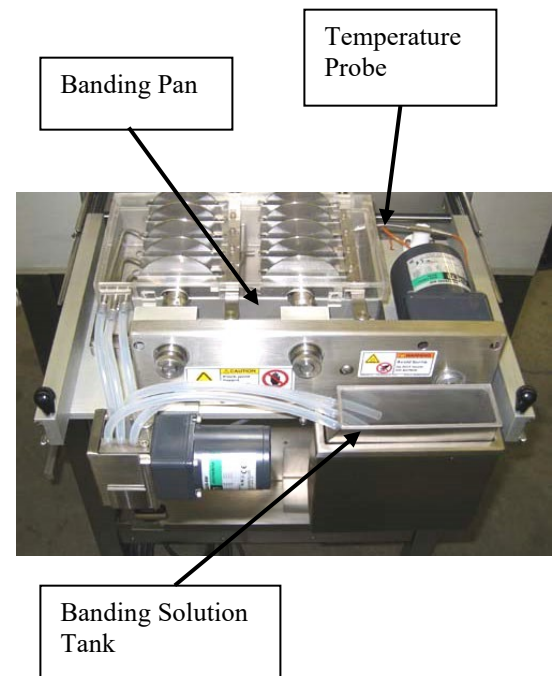
- 3.1.1. Prepare the solution based on manufacturers instructions in the following quantities:

For banding pan 700g (350g for each 1st and 2nd seal pan)

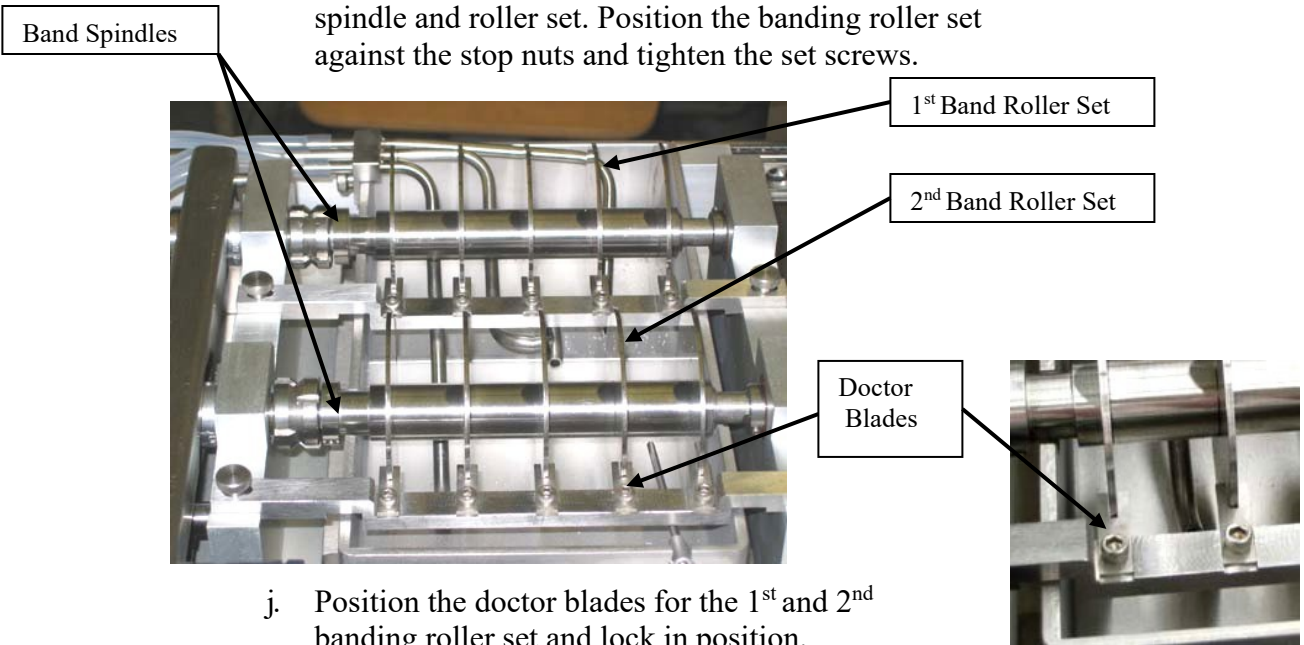
For banding tank 800g

- 3.1.2. Assemble the band unit according to the following procedure:

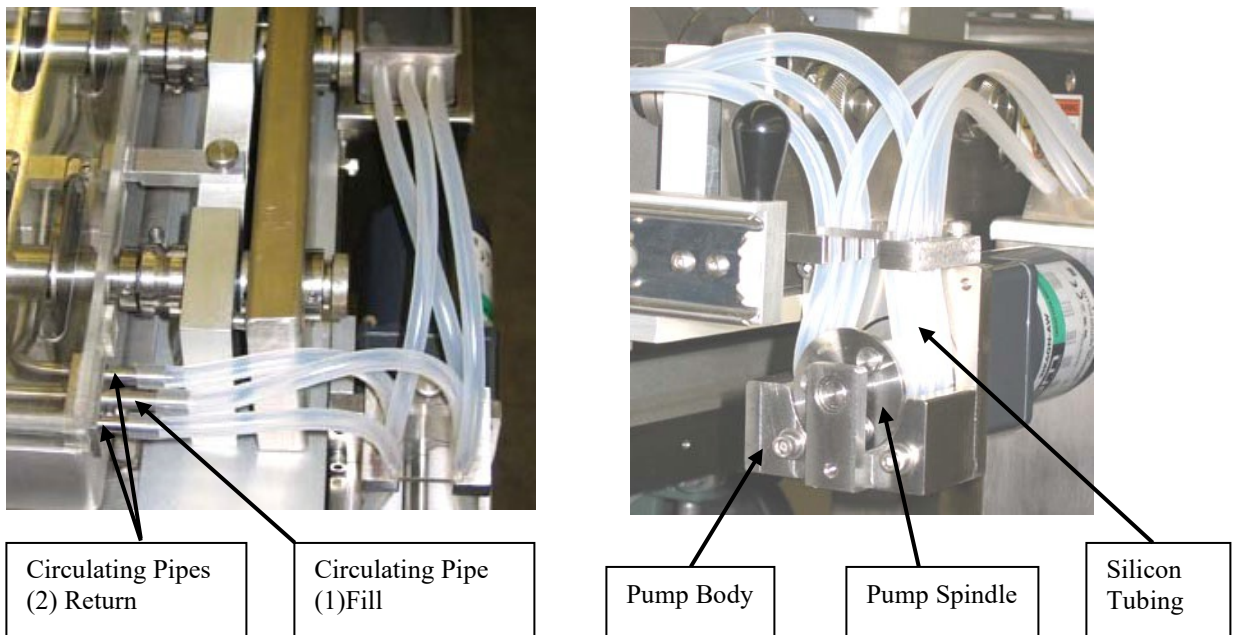
- a. Open banding unit door.
- b. Pull the banding drawer assembly out.
(must be in lowered position)
- c. Place the banding pan in position.
- d. Place the banding solution tank in position.
- e. Attach the temperature probe on the right side of banding pan.
- f. Verify that the temperature of the banding solution controller is set at the required setting.
- g. Place the circulating pipes in the in the left side of the banding pan, and tighten the roller knob.



- h. Set 1st and 2nd banding rollers sets in position between the spindle shaft bearings.
- i. Insert the spindle shaft through the banding spindle and roller set. Position the banding roller set against the stop nuts and tighten the set screws.

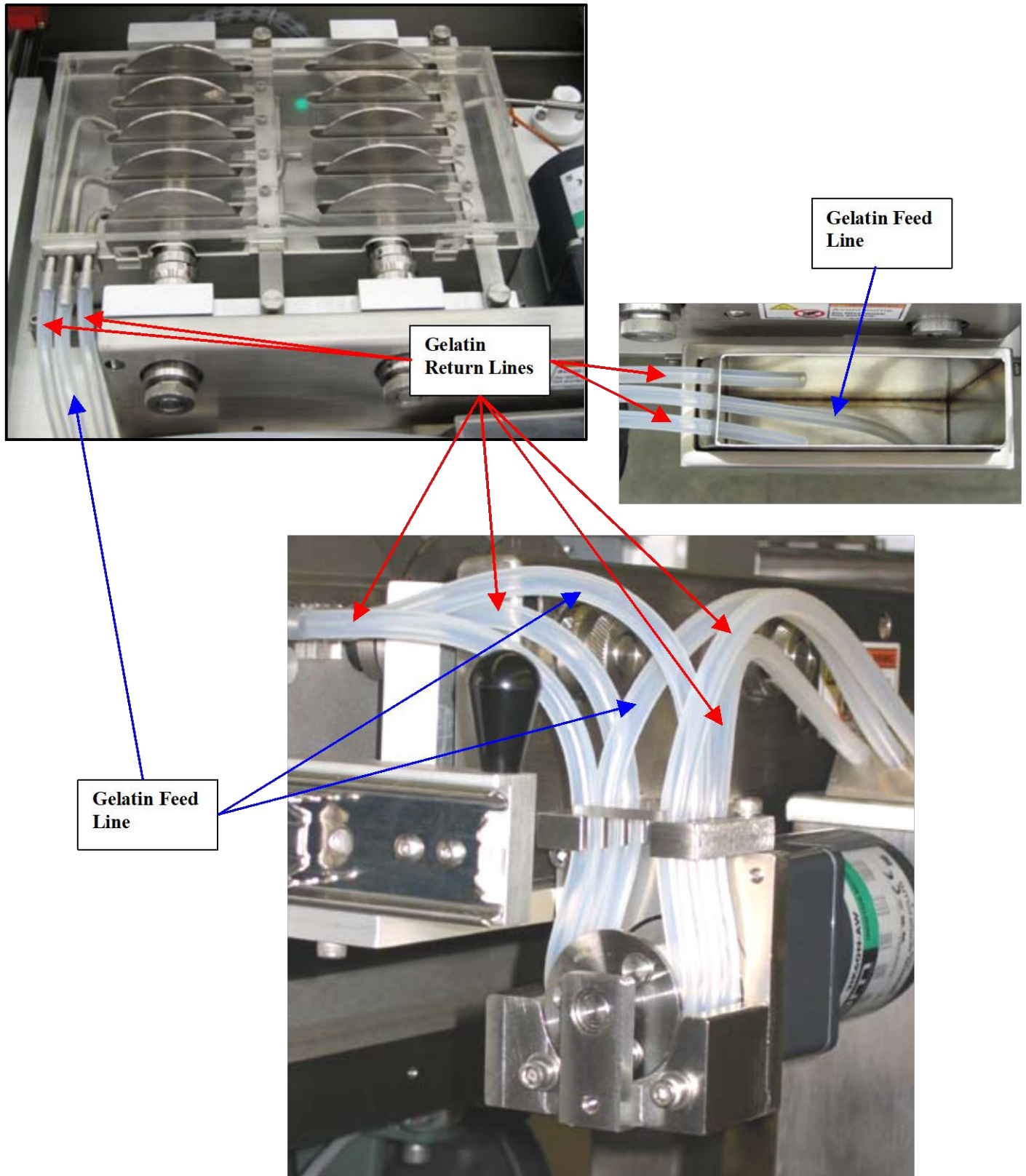


- j. Position the doctor blades for the 1st and 2nd banding roller set and lock in position. Check the for proper clearance with the banding rollers.
- k. Assemble the silicon tubing from the circulating pipes around the pump spindle and into the banding solution tank.

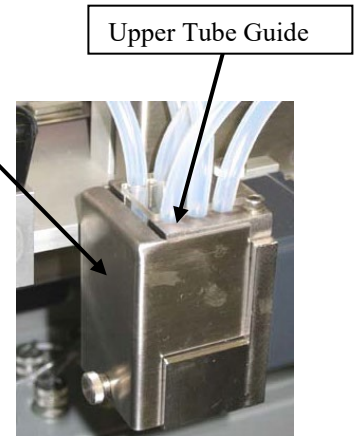


STI CB-15

ROUTING INSTRUCTIONS for GELATIN FEED and RETURN TUBING



- l. Push the Pump Body toward the motor and Secure in place (silicon oil may be used inside the pump body).
- m. Attach the upper tube guide cover and secure in place.
Attach the pump body cover and secure in place.
- n. Position the silicon tubing into the banding solution tank as shown. (note: the suction tube must extend to the bottom of the tank.)



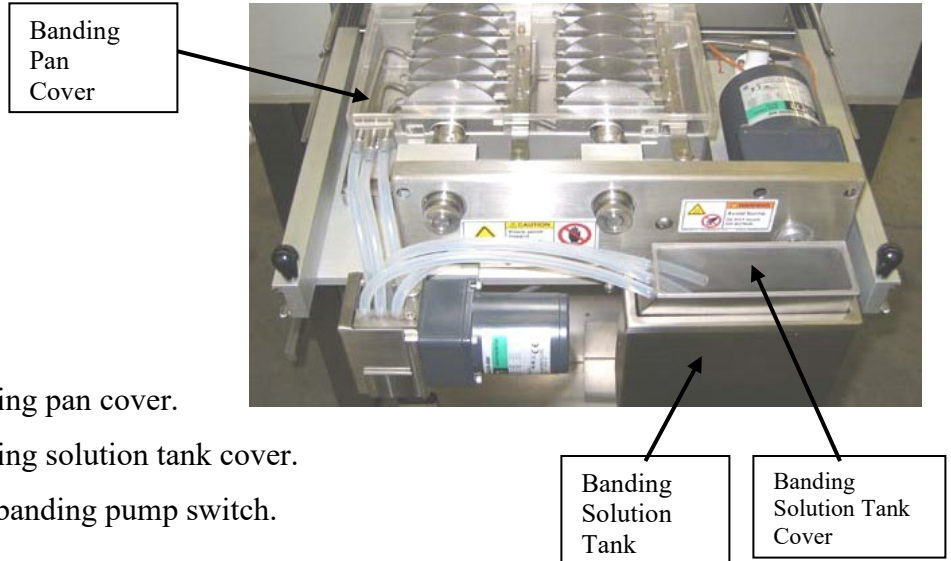
3.1.3. Turn main electrical switch “ON,” and supply the compressed air and vacuum.

3.1.4. Check for remaining capsules:
Turn “ON” main drive switch and perform a blank run operation to check that there are no capsules remaining. Check the capsule pull-off unit as well.

4. Production

4.1.1. Fill banding pan and the banding solution tank respectively:

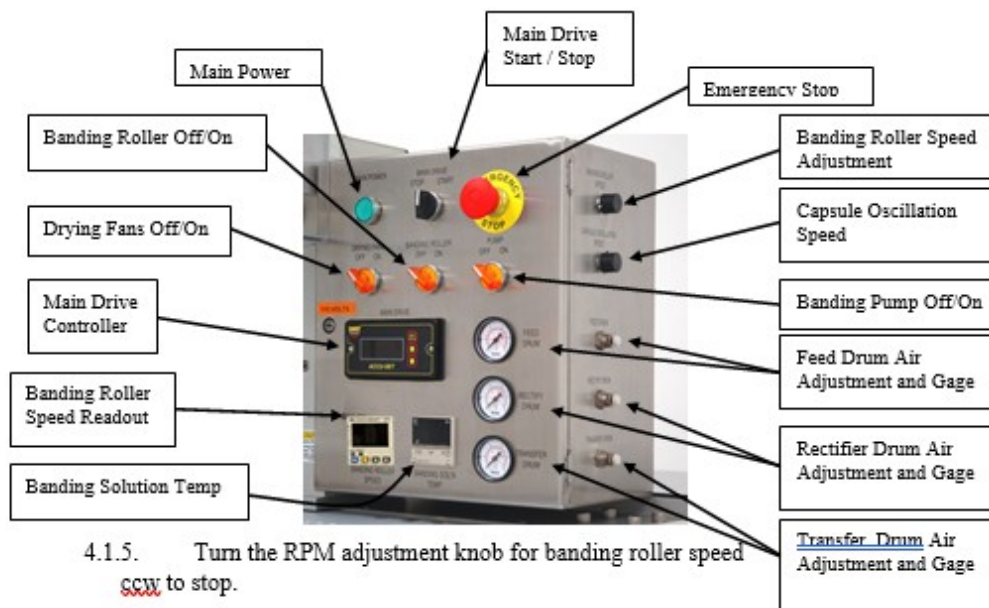
Banding pan 1st seal..... 350g
 2nd seal..... 350g
 Banding tank...800g



4.1.2. Attach banding pan cover.

4.1.3. Attach banding solution tank cover.

4.1.4. Turn “ON” banding pump switch.



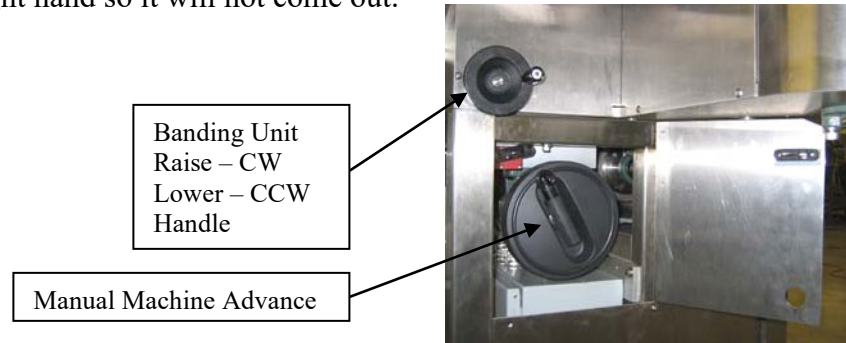
4.1.5. Turn the RPM adjustment knob for banding roller speed ~~ccw~~ to stop.

4.1.6. Turn “ON” banding roller switch.

4.1.7. Adjust to 150rpm by turning RPM adjustment knob.

4.1.8. Open the cover of banding solution tank, and be sure that the solution is running from both return tubes equally.

- 4.1.9. Slide the banding drawer assembly into the run / banding position.
- 4.1.10. Rotate the handle for banding unit to right to lift the unit to a suitable position. At this time, support banding unit with right hand so it will not come out.

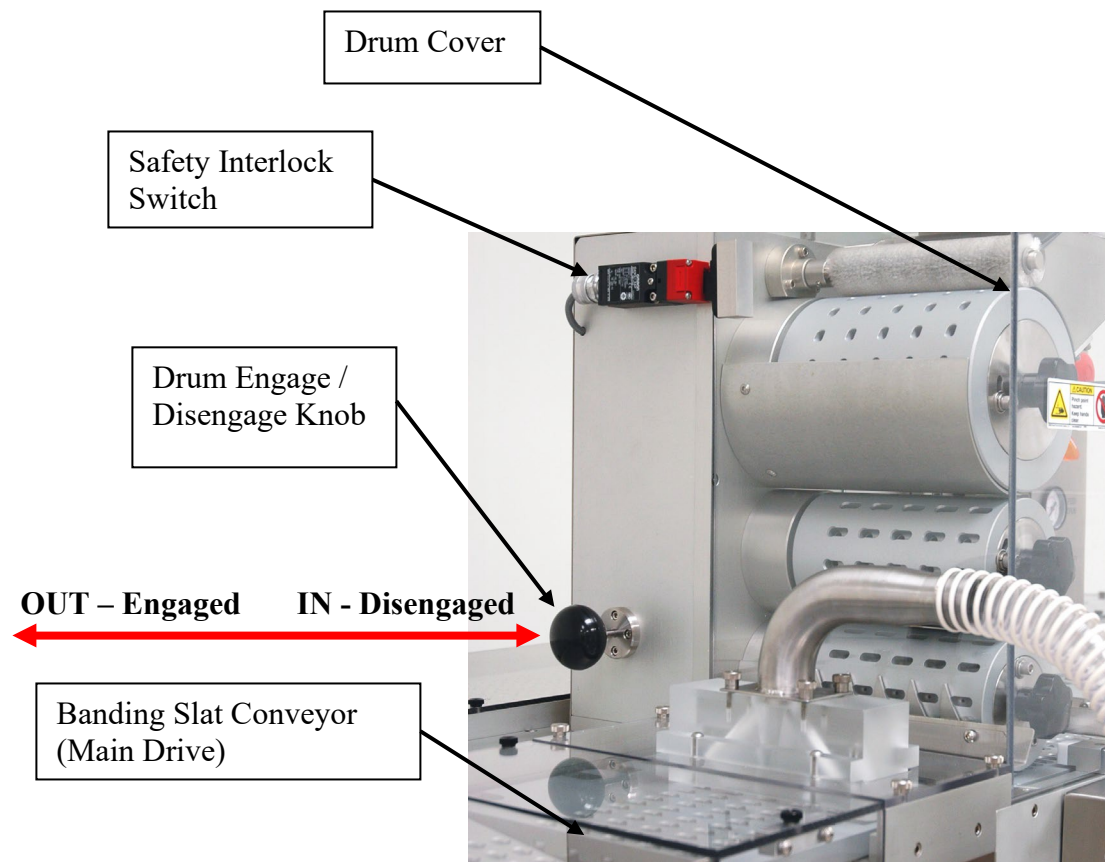


- 4.1.11. Adjust the height of banding unit.
- 4.1.12. Shut banding unit door.
- 4.1.13. Supply 10~20 filled capsules into hopper.
- 4.1.14. Turn “ON” main drive switch and fan.
- 4.1.15. Watch for errors during operation, making adjustments as needed.
- 4.1.16. Turn “OFF” main drive switch.
- 4.1.17. Be sure that rectification, banding and drying are going well, then insert filled capsules into hopper.
 Note: Check maximum leakage time after filling of low low viscosity liquid filling capsules. Banding should be finished within the time limited. Leakage problems would be found 30 minutes after filling with the viscosity 30cps/25°C.
- 4.1.18. RPM of Banding roller:
- | | |
|----------------|------------|
| At production | 150~200rpm |
| At standing-by | 150rpm |

4.1.19. Turn “ON” main drive switch.

Note: At any time during the machine cycle the feed and rectifying drums may be disengaged by pushing in the large black knob located on the side of the drum drive assembly. The clear Lexan drum cover can then be removed without stopping the main slat drive or banding rollers. (If the drum drive is in the engaged position and the drum cover is removed – ALL MACHINE FUNCTIONS ARE SHUT DOWN and the machine will have to be restarted.

To Re-engage the drum drive – re-install the clear Lexan drum cover and then pull the black knob out (away) from the drum drive assembly. There is a single pickup point for the drums to engage. From the time the knob is pulled out there will be a potential lag time of from zero to 30 seconds before the drums start to rotate.



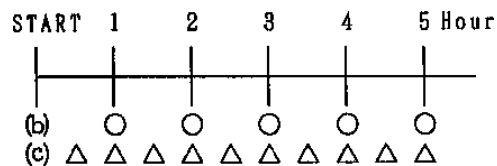
4.1.20. Work during operation

- i. Supply filled capsules.
- ii. Supply banding solution.
To keep the quantity of solution in banding tank, replenish the decreasing quantity once an hour.
- iii. Supply 80cc of hot deoxidized water (50-55°C, depending on the gelatin's operating temperature.)
For banding tank once every 60 minutes – add gelatin as needed.
Evaporation Rate:

4.1.20 Work during operation (Continued)

- Measure/mark the gelatin level in the pan. Run 1 hour at idle (bring to temperature and then circulate.)
- Measure the amount of water required to raise the level back to the original point. This amount will need to be added every hour to compensate for evaporation. (Retest the evaporation rate whenever the environment or color of the gelatin dye is changed.)

Note: After supplying, stir the solution in banding tank.



- iv. Replace the capsule tray on capsule take-off unit.
- v. Check on the banding conditions.

4.1.21. After production, perform a blank operation to check for capsule remains.

4.1.22. Turn “OFF” switches on the fan and Main drive.

4.2. Disassembling and washing

- 4.2.1. Open banding door.
- 4.2.2. Turn the handle for banding unit (ccw) to left to lower the unit to a suitable position.
- 4.2.3. Pull out banding unit slowly.
- 4.2.4. Turn the RPM adjustment knob for banding roller to stop.
- 4.2.5. Turn “OFF” banding roller switch.
- 4.2.6. Turn “OFF” banding pump switch.
- 4.2.7. Remove banding pan cover and banding tank cover.
- 4.2.8. Loosen the set bolt of each 1st and 2nd banding roller and spindle, and pull the spindle out.
- 4.2.9. Remove doctor blade assemblies.
- 4.2.10. Take the removed parts out for washing.
- 4.2.11. Remove the temperature probe and wipe with cloth.
- 4.2.12. Disassemble the banding solution circulating pump.
- 4.2.13. Remove the solution circulating pipes.
- 4.2.14. Take the removed banding pan and banding tank out for washing.
- 4.2.15. Push into banding unit slowly, and shut banding door.
- 4.2.16. Wash the removed parts with hot water (over 40°C) after leaving in hot water for awhile, and then dry.
Note: Be sure that there is not any damage on silicon tubing, and replace if needed.

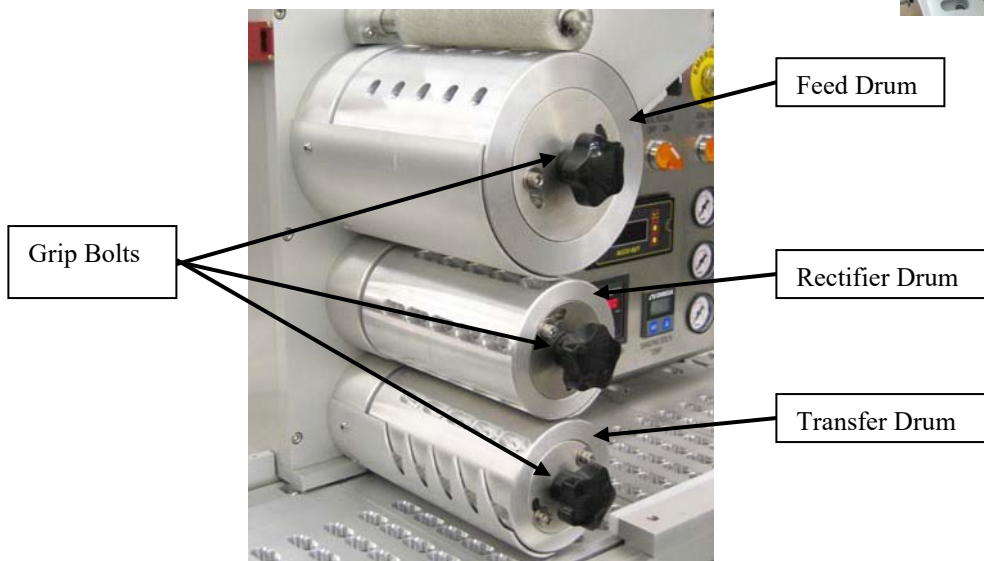
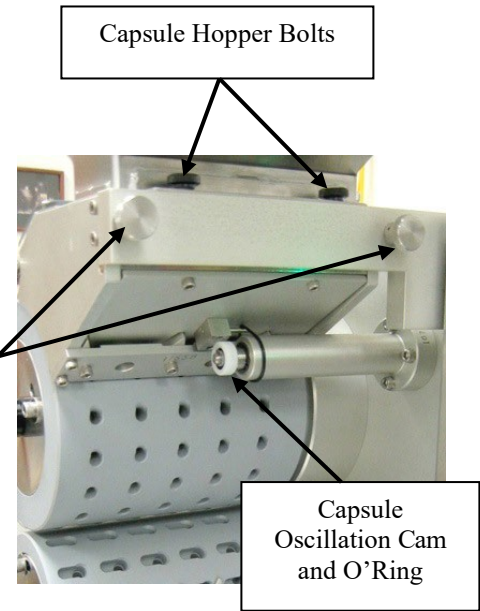
4.0 Size Change

4.1 Disassemble the necessary parts.

4.1.1 Remove rectifier assembly cover.

4.1.2 Remove the bolts for the capsule oscillator and hopper and remove the capsule hopper.

4.1.3 Remove the grip bolt for attaching the feed drum.



4.1.4 Remove the grip bolt for attaching the rectifier drum.

4.1.5 Remove the grip bolt for attaching the transfer drum.

4.1.6 Remove feed drum.

4.1.7 Remove rectifier drum.

4.1.8 Remove transfer drum.

- 4.1.9 Loosen the bolts for attaching the slats and take the slats out at this time.

Slat Removal Bolts

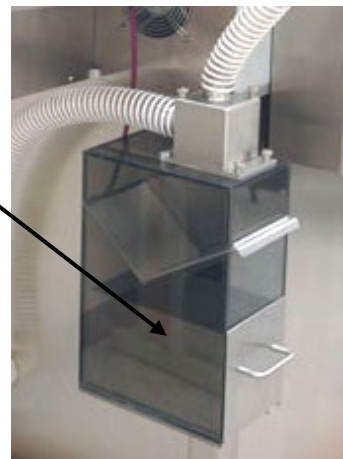


Capsule Pull-off Block



- 4.1.10 Remove the capsule pull-off block and the collection box.

Collection Box



- 4.2 Assemble the change parts according to the following procedure.
 - 4.2.1 Attach the suitable size slats. Be sure that the extremity of slat is touched with bracket.
 - 4.2.2 Attach transfer drum and tighten the grip bolt.
 - 4.2.3 Attach rectifier drum and tighten the grip bolt.
 - 4.2.4 Attach feed drum and tighten the grip bolt.
 - 4.2.5 Attach appropriate size hopper assembly.
 - 4.2.6 Attach the capsule hopper with the mounting bolts.
 - 4.2.7 Attach the rectifier unit cover.
 - 4.2.8 Attach the capsule pull-off block, hose and collection box.

5. Trouble shooting

If an error occurs during operation, resolve using the following as a guide.

If you continue to have problems with the machine, please contact the Schaefer Technologies, Inc. sales department.

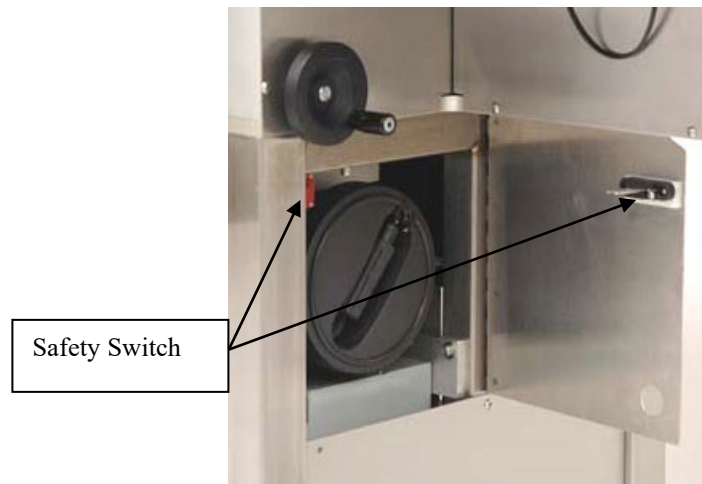
Problem #1: When pushing main power switch “ON,” lamp does not light.

- Cause:
- 1) Machine is not connected to electrical source of your plant.
 - 2) Main breaker is switched “OFF.”
 - 3) Circuit protector is working.
 - 4) Drying Fan, Seal Roller and Seal Pump is in “ON” position.

- Corrections:
- 1) Connect to the electrical source and turn “ON” the breaker of your plant.
 - 2) Turn main breaker “ON.”
 - 3) Confirm that there isn’t any trouble in main wiring.
 - 4) Turn Drying Fan, Seal Roller and Seal Pump to “OFF” position.

Problem #2: When pushing main drive switch “ON,” machine does not start running.

- Cause:
- 1) Door safety switches are not made (3) total.



- Corrections:
- 1) Close all doors / install all interlocked covers.

Problem #3: Capsules are not feeding from capsule hopper to feed drum or a low feed ratio.

- Cause:
- 1) Capsule quantity is too low in hopper.
 - 2) Oscillation plate speed too slow.
 - 3) Vacuum not turned on.

- Correction:
- 1) Supply capsules.
 - 2) Increase oscillation plate speed.

Problem #4: Capsule sticks in cavity of feed drum, rectifier drum, or Transfer drum.

- Cause:
- 1) Defective capsule stuck in cavity.
 - 2) Too low air flow.
 - 3) Leakage from defective capsules.

- Correction:
- 1) Remove defective capsule and wipe with cloth moistened with ethylalcohol.
 - 2) Adjust the air flow valve.
 - 3) Remove and wipe with cloth moistened with ethylalcohol.

Problem #5: Capsules are coming through the brush roller along with the rotation of the feed drum.

- Cause:
- 1) Too many capsules in capsule hopper.
 - 2) Slow rotation of brush roller.

- Correction:
- 1) Reduce the capsule quantity.
 - 2) Increase speed of brush roller.

Problem #6: Transfer to cavities for each drum and/or to slat is unsatisfactory.

- Cause:
- 1) Drum alignment not correct.

- Correction:
- 1) Realign drums.

Problem #7: Capsules are not taken off the slat.

- Cause:
- 1) Low pressure / vacuum of the air for capsule pull-off unit.
 - 2) Contamination in the slat pockets.

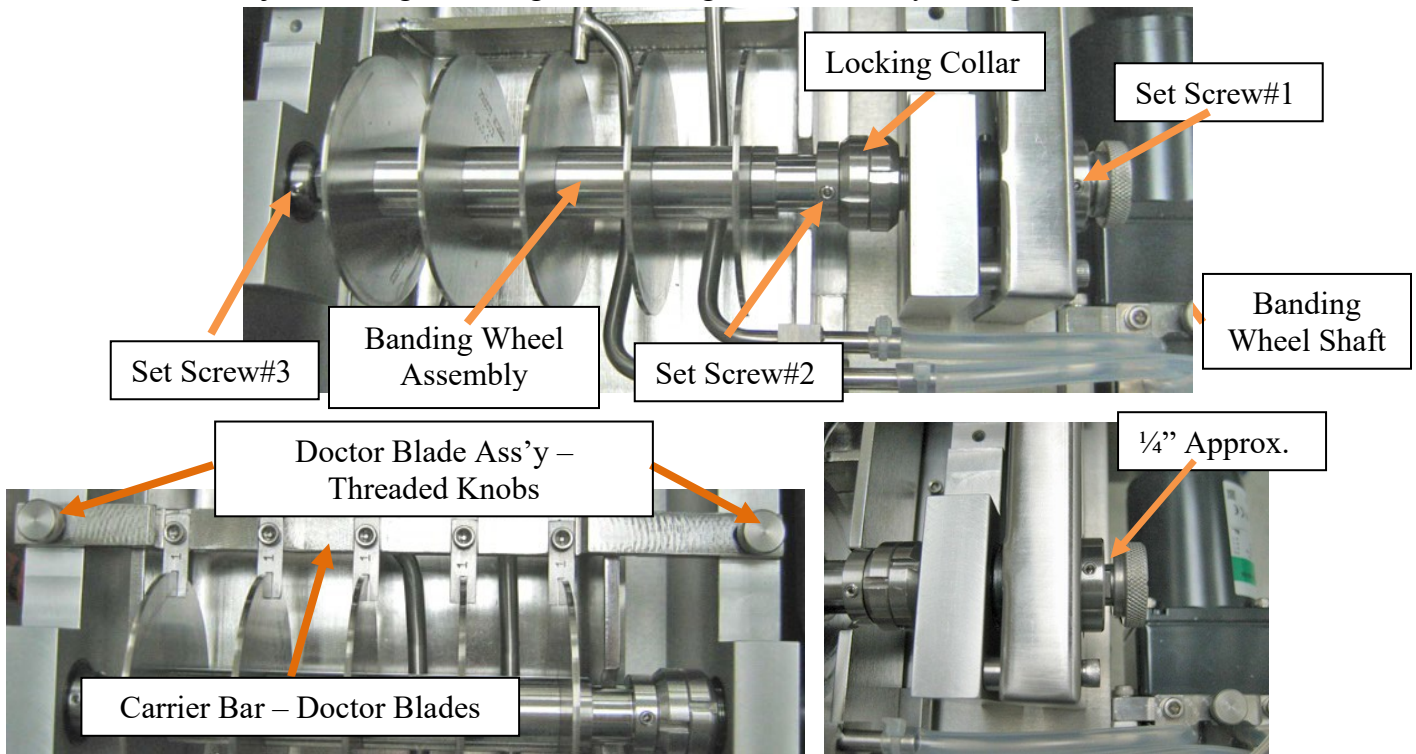
- Correction:
- 1) Increase air pressure / vacuum to capsule pull-off.
 - 2) Clean the slat.

Problem #8: Banding wheel out of alignment with capsule body/cap seam.

Correction: Re-align Banding Wheel Assembly

Procedure: (Reference photos below)

1. Remove (2) knurled knobs and doctor blades ass'y. Loosen set screws #1, #2, & #3. Pull shaft out approx. $\frac{1}{4}$ " and slide banding wheel ass'y against locking collar and tighten set screw #2. Loosen locking collars (1) turn – do not tighten.
2. Push in tray assembly – hold in position and rotate banding unit handle CW to raise the banding unit until locating pin engages. Insert assembled capsule into banding slot (cap end away from operator) and position slot so the approx. center of the banding wheel is below the capsule centerline. Raise the banding unit until contact is made with the capsule. By hand - rotate banding wheels 3-4 revolutions to position capsule in slot. Pushing banding wheel shaft in or out, center banding wheel on the capsule cap/body seam. Tighten set screw #1.
3. Rotate banding unit handle CCW to lower assembly. Pull out banding assemble tray. Adjust locking collars against banding wheel assembly and tighten.



4. Verify Alignment of banding wheel to capsule cap/body seam by pushing in tray assembly – hold in position and rotate banding unit handle CW to raise the banding unit until locating pin engages. Insert assembled capsule into banding slot (cap end away from operator) and position slot so the approx. center of the banding wheel is below the capsule centerline. Raise the banding unit until contact is made with the capsule – rotate banding wheels 3-4 revolutions to position capsule in slot.
5. Rotate banding unit handle CCW to lower assembly. Pull out banding assemble tray. Loosen set screws #1, #2 and #3. Push banding wheel shaft in fully and tighten Set screws #1 and #3. Make sure banding wheel assembly is against the locking collars and tighten set screw #2. Re-attach doctor blade assembly with knurled knobs.

CB-15 Recommended Spare Parts

Part Number	DESCRIPTION	REQ
CB-1380	Banding Wheel – First Stage “V” (Sizes / 000,00,0,1)	5
CB-1381	Banding Wheel – First Stage “V” (Sizes / 2,3,4)	5
CB-1390	Banding Wheel – Second Stage “Flat” (Sizes / 000,00,0,1)	5
CB-1391	Banding Wheel – Second Stage “Flat” (Sizes / 2,3,4)	5
CB-1047	Doctor Blades (Sizes / 000,00,1)	10
CB-1047A	Doctor Blades (Sizes / 2,3,4)	10
CB-1628	Pharma-50 Tubing – 50ft.	1
CB-1164	Brush Assembly	1
CB-1513A	Drying Fan Filter Assembly	7
CB-3034	Heating Element – 75watt	1
CB-3035	Heating Element – 330watt	1
CB-1400-07	Gelatin Pump Retainer Cap	1
CB-1400-10	Gelatin Pump Bearing	1
CB-1052	Rail Link	40
CB-1154A	Capsule Stop Wiper	1
CB-2037	O-Ring	2
CB-1400-04	Pump Roller	3
CB-1400-12	Gelatin Pump Bearing for Rollers	6
CB-1400-13	Waldes E-Ring	3
CB-1212	Vacuum Plate - Feed	1
CB-1213	Vacuum Plate – Rectify	1
CB-1214	Vacuum Plate – Transfer	1
CB-1605	Spring for Vacuum Plate	12
CB-1526	Bearing (Hopper Finger & Brush Spindle)	4
CB-1101	Guide Rail (Wear Strip)	1
CB-1110	Guide Rail	1
CB-1101A	Guide Rail (Wear Strip Extension)	1
CB-2802	Vacumm Pump Canister Filter	2
CB-1039	Tubing Assembly	1
CF-2729	Fuse – 5 Amp	1
CF-2785	Fuse – 15 Amp	1
CF-2731A	Fuse – 1 Amp	1

Optional Vacuum Pump & Filter Canister

